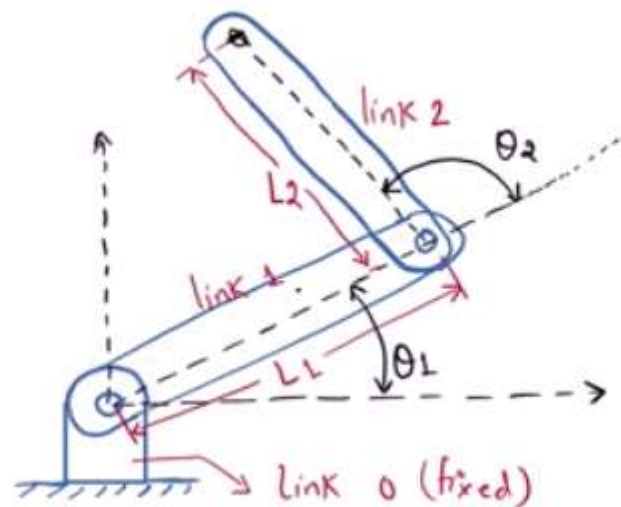


Aim: To study reverse Kinematics of 2 dof planer serial manipulator

Instrument required: Desktop PC with internet connection.

Inverse Kinematics:

Inverse kinematics (IK) in robotics is the mathematical process of calculating required joint angles and parameters to position a robot's end-effector (gripper, tool) at a specific Cartesian coordinate and orientation. It is the reverse of forward kinematics, allowing robots to move to desired locations



Derive the closed form IK of this 2 Dof manipulator.

Given Equation:

$${}^0T_2 = {}^0T_1 \times {}^1T_2$$

Configuration Matrix:

$$T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & P_x \\ r_{21} & r_{22} & r_{23} & P_y \\ r_{31} & r_{32} & r_{33} & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Configuration matrix for above manipulator for $\theta_1 = 0$ and $\theta_2 = 0$

$$T = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Equating general equation with manipulator equation:

$$P_x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$P_y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

Solve for theta2:

$$P_x^2 + P_y^2 = L_1^2 + L_2^2 + 2L_1L_2 \cos \theta_2$$

$$\cos \theta_2 = \frac{P_x^2 + P_y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

$$\theta_2 = \cos^{-1} \left(\frac{P_x^2 + P_y^2 - L_1^2 - L_2^2}{2L_1L_2} \right)$$

For theta1:

$$\theta_1 = \tan^{-1} \left(\frac{P_y}{P_x} \right) - \tan^{-1} \left(\frac{L_2 \sin \theta_2}{L_1 + L_2 \cos \theta_2} \right)$$

Matlab Code:

```
clc;
clear;
Px = 9.1416; % end effector x
Py = 14.7984; % end effector y
L1 = 10; % link1 length
L2 = 8; % link2 length

D = (Px^2 + Py^2 - L1^2 - L2^2)/(2*L1*L2);

theta2 = atan2(sqrt(1-D^2), D);

theta1 = atan2(Py, Px) - atan2(L2*sin(theta2), L1 + L2*cos(theta2));

disp('Theta1 =')
disp(theta1)

disp('Theta2 =')
disp(theta2)
```

Answer:

Theta1 =
0.7854

Theta2 =
0.5236

Result:

The screenshot shows the MATLAB IDE interface. The top toolbar includes buttons for 'Bookmark', 'Debugger', 'Run Section', 'Run to End', 'Run', 'Step', and 'Stop'. Below the toolbar, the 'Run' button is highlighted. The main workspace displays the script 'inverse1.m' with the following code:

```
6 L2 = 8; % link2 length
7
8 D = (Px^2 + Py^2 - L1^2 - L2^2)/(2*L1*L2);
9
10 theta2 = atan2(sqrt(1-D^2), D);
11
12 theta1 = atan2(Py, Px) - atan2(L2*sin(theta2), L1 + L2*cos(theta2));
13
14 disp('Theta1 =')
15 disp(theta1)
16
17 disp('Theta2 =')
18 disp(theta2)
```

The 'Command Window' at the bottom shows the output of the script:

```
Theta1 =
0.7854
Theta2 =
0.5236
>>
```